

NovaMart

Multi-Vendor E-Commerce Marketplace

UI/UX Design Case Study

Vendor Onboarding • Smart Search • Seamless Checkout • Real-Time Tracking

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Screens	13 hi-fi screens (mobile + desktop)
Prototype	68 interactive connections across 4 user flows

“Poor UX in e-commerce causes 70%+ cart abandonment. NovaMart delivers an intuitive, conversion-focused marketplace experience with smooth vendor onboarding, smart search, seamless checkout, and delightful interactions – designed to boost conversion rates by 25–40% and create memorable user journeys.”

1. Project Overview

Vision & Mission

Design a modern, user-centric multi-vendor e-commerce marketplace that provides delightful experiences for three key user types: **Customers**, **Vendors**, and **Admins**. The focus is on reducing friction, increasing trust, and maximising conversions through thoughtful information architecture, micro-interactions, and accessibility-first design.

Business Value Delivered

Target Users

#	Persona	Role	Primary Goal on NovaMart
1	Priya Sharma	Customer — Marketing Mgr	Fast visual discovery, no sponsored clutter
2	Arjun Mehta	Customer — Student	Spec comparison, counterfeit protection
3	Sunita Kapoor	Customer — Homemaker	Simple navigation, 1-tap reorder
4	Rohit Desai	Customer — UX Designer	Design quality, dark mode, vendor stories
5	Vinod Nair	Customer — Retired	Accessible typography, transparent pricing
6	Kavya Reddy	Vendor — Home Reseller	Brand identity, simple mobile listing
7	Sameer Joshi	Vendor — Electronics SMB	Analytics dashboard, inventory alerts
8	Meena Iyer	Vendor — Artisan Seller	Craft storytelling, custom order flow

Non-Functional Design Requirements

- Mobile-first responsive design
- WCAG 2.1 AA accessibility
- 60fps animations
- Dark / Light mode support
- Inclusive design (diverse personas)
- ≤3 taps to any primary action

2. Key Features & User Flows

ID	Feature	Description & Design Value	Key Deliverables
F01	Vendor Onboarding	4-step wizard with progress bar, time estimate per step, Save & Continue Later. Reduces Amazon's 12-step process to 4.	Onboarding screens, progress component, success celebration
F02	Smart Product Discovery	Style-based filter chips, visual/camera search, zero sponsored results in top 3. Personalized recommendations.	Search UI, filter system, product cards
F03	Cart & Checkout	Persistent cart drawer (no page redirect). Single-page checkout with UPI default, trust row above Pay button.	Cart drawer, checkout flow, payment UI
F04	Order Tracking	Live map with agent pin, 5-step timeline, proactive push notification 24hrs before delivery.	Tracking page, timeline component, push alert
F05	Vendor Dashboard	Revenue charts, conversion funnel, traffic sources, low-stock alerts. Replaces Meesho's absent analytics.	Vendor portal UI, charts, tables
F06	Admin Panel	Vendor KYC approval queue, content flagging, dispute resolution, platform GMV dashboard.	Admin panel with data tables, approval queue

3. Design Process



3.1 Research & Discovery (Days 1–4)

Competitor Analysis — Evaluated Amazon, Flipkart, Meesho, and Etsy across vendor onboarding, search UX, checkout flow, and trust mechanisms.

Key findings that shaped NovaMart’s design:

- Amazon requires 12+ steps for vendor registration with a 2-day approval wait
- Flipkart’s top 3 search positions are always sponsored — organic results start at position 4
- Meesho gives vendors zero brand identity — all stores look identical
- Etsy has no visual/image search despite a highly visual product catalogue
- All four platforms use multi-step checkout — none offers single-page on mobile

User Research — Created 8 personas through research synthesis: 5 customer personas covering urban professionals, students, homemakers, designers, and senior first-time shoppers; 3 vendor personas covering home resellers, SMB electronics retailers, and artisan craft sellers.

Journey Mapping — Mapped 3 end-to-end journeys: J1 (Customer Shopping — 6 stages), J2 (Vendor Onboarding — 5 stages), J3 (Order Tracking — 5 stages). Identified 15 distinct pain points across all three journeys.

Information Architecture — Defined 54 screens across 3 portals, 5-tab bottom navigation (all primary actions in 1 tap), vendor sidebar navigation, and P1/P2/P3 priority tiers for development.

3.2 Design System (Day 8)

Built before any screen was designed — establishing the source of truth for all visual decisions:

- **Portal colour identity:** Customer (#4F46E5 purple), Vendor (#0F766E teal), Admin (#193B6D navy)
- **Typography:** Inter, 8-step scale from Display/H1 (32px) to Label/Small (12px)
- **Spacing scale:** 8-token system from 4px to 64px
- **8 core components:** Button, Input Field, Product Card, Bottom Nav, Top App Bar, Badge/Chip, Rating Stars, Toast

3.3 High-Fidelity Screens (Days 9–13)

Designed 13 P1 screens mobile-first (390×844px), followed by 6 desktop adaptations (1440×900px). Each screen was built using the shared design system tokens — no manual colour or font picking.

3.4 Prototyping (Days 15–17)

Created 68 prototype connections across 4 flows using a custom Figma plugin (NovaMart Wire Prototype) that automates hotspot creation and applies dissolve transitions at 300ms ease-out. Three user journeys are fully clickable: J1 Shopping (14 connections), J2 Vendor Onboarding (8 connections), J3 Order Tracking (5 connections), plus 40 sticky bottom nav connections across all 8 customer screens.

4. Design System Summary

Category	Token	Value & Usage
Customer CTA	Primary/Brand	#4F46E5 — buttons, active nav, hero banner
Vendor CTA	Vendor/Teal	#0F766E — vendor buttons, progress bar
Admin Sidebar	Admin/Navy	#193B6D — sidebar, header
Admin Alert	Admin/Coral	#DC4E3A — flagged content, reject
Success / Live	Status/Green	#15803D — delivered, active step
Warning / Pending	Status/Amber	#B45309 — processing, pending
Body Text	Text/Dark	#374151 — all body copy
Font family	Inter	Google Fonts — 8-step scale 12px–32px
Base spacing	md token	16px — standard screen padding
Card radius	8px	Cards, inputs, drawers

5. Technical Highlights

5.1 Micro-Interactions Designed

Interaction	Description	Duration
Add to Cart	Product image flies to cart icon with spring animation	400ms
Cart Drawer	Bottom sheet slides up over dimmed background	300ms
Order Timeline	Active step pulses with dot animation	Loop
Go Live Celebration	Animated checkmark + confetti on vendor store launch	600ms
Order Confirmation	Checkmark draws in on success screen	500ms
Filter Chip Select	Chip fills with Primary/Light on selection	150ms
Push Notification	Slides in from top, auto-dismisses	250ms

5.2 Accessibility Implementation

- All text colour combinations verified at WCAG 2.1 AA (4.5:1 for normal text, 3:1 for large)
- Minimum touch target size 44×44px on all interactive elements
- Bottom nav uses icon + text label (never icon-only)
- Font size toggle in Settings screen — addresses senior users (Vinod persona)
- Status indicators use icon + colour + text label (never colour alone)
- Flipkart's WCAG-failing red sale labels (#DC2626 on white = 3.9:1, fails AA) corrected in NovaMart to Status/Amber at 4.7:1

5.3 Challenges Faced & How Solved

Challenge	Solution
Balancing simplicity for first-time shoppers (Sunita/Vinod) vs power features for expert users (Arjun/Rohit)	Progressive disclosure — simple default view with collapsible advanced sections on Search and PDP
Consistent cross-portal experience while maintaining distinct vendor vs customer visual identity	Shared design system tokens for typography/spacing/radius; portal-specific accent colour only
Vendor onboarding on mobile with complex KYC requirements	4-step wizard with time estimate, inline validation, Save & Continue Later — reduces cognitive load per step
Multi-vendor cart creating separate checkout per vendor (Etsy's core pain point)	Aggregated cart drawer with combined price breakdown and single Pay button regardless of vendor count

6. Key Design Decisions

Each decision below is traced directly to a competitor gap or persona pain point from the Day 1–2 research.

Pain: Amazon and Flipkart both place sponsored results in positions 1–4, creating banner blindness and eroding trust (Priya, Day 1 competitor analysis).
Fix: NovaMart shows zero sponsored results in the top 3 positions. Organic results appear first. Sponsored content (if any) is clearly labelled and appears below position 4.

Pain: All 4 competitors use multi-step checkout on mobile (3–12 steps). This is the single largest cause of cart abandonment across the industry.
Fix: Single-page checkout: delivery address, slot, payment method, order summary, price breakdown, trust row, and Pay button on one scrollable screen. UPI is pre-selected as default payment.

Pain: Return policy is buried 3+ levels deep on Amazon and Flipkart. Priya abandons carts specifically because she cannot quickly verify return terms (discovered in journey mapping J1).
Fix: Vendor trust card on every PDP: vendor name, rating, response time, and return policy in a single visible card — no tapping required.

Pain: Meesho gives vendors zero brand identity — every store looks identical. Kavya explicitly wants customers to remember her brand, not just her products (Vendor persona Day 2).
Fix: Vendor storefront with customisable cover photo, logo, colour accent, bio, and Instagram link. Live preview panel shows how it appears to customers before saving.

Pain: Sunita cannot find “My Orders” on any competitor app without help — it is buried under Profile → Account → Purchases (journey map J3, emotion dip stage 2).
Fix: “Orders” is a dedicated bottom nav tab — 1 tap from any customer screen. Order cards show product image and status chip without any additional navigation.

7. Usability Testing & Iterations

Test Setup

- **Round 1:** Wireframe testing on Day 6 using Maze
- **Round 2:** Hi-fi prototype testing on Day 19
- **Final round:** 8-participant test on Day 22
- **Profiles:** 5 customer users, 3 vendor users matching persona demographics

Key Findings & Iterations

#	Finding	Iteration Made	Outcome
1	3 of 5 customers missed the camera search icon — it was too small and low-contrast	Increased camera icon size to 24px, added #4F46E5 tint	100% discovery in re-test
2	Vendor users (Kavya persona type) confused by purple accent on V01–V04 onboarding — expected same teal as the rest of the vendor portal	Swapped V01–V04 progress bar, Next, and Go Live button from #4F46E5 to #0F766E	Visual consistency confirmed
3	Senior user (Vinod persona type) didn't notice the “7-day returns” text in the trust row — too small	Increased trust row text from 11px to 13px, added border around trust row	Read by all participants
4	Cart drawer drag handle not intuitive for first-time users	Added “Swipe down to close” helper text beneath drag handle	Dismiss rate improved
5	Vendor analytics screen (V12) overwhelming on first view — too many data points	Added “Last 30 days” as default, other ranges collapsed behind date picker	Task completion up

8. Execution Timeline

Phase	Days	Key Activities	Deliverable
Research	1–4	Competitor analysis (4 platforms), 8 personas, 3 journey maps, 54-screen sitemap, navigation hierarchy	4 research documents
UI Design	5–14	Design system (17 colour tokens, 8 text styles, 8 components), 13 hi-fi screens mobile, 6 desktop adaptations	Figma design file
Prototyping	15–21	68 prototype connections, micro-interaction specs, dark mode implementation, WCAG audit	Interactive prototype
Testing	22–23	8-participant usability test, 5 iterations based on findings	Test report
Handoff	24–25	Design system documentation, component library, Figma Dev Mode specs	Design system PDF
Presentation	26–28	Case study writing, demo video recording, final QA, portfolio upload	This document + video

9. Personal Reflection

Key Learnings

Research before pixels. The 4 days spent on research before opening Figma paid dividends across every design decision. Every screen has a traceable reason for existing — a persona pain point, a journey map gap, or a competitor weakness. Without that foundation, the 13 screens would have been visually polished but strategically hollow.

Design systems are the real deliverable. The 8 personas, 3 journey maps, and 54-screen sitemap are each valuable artifacts. But the design system — 17 colour tokens, 8 type styles, 8 components — is what made 13 screens look like one coherent product rather than 13 individual efforts.

Portal colour identity emerged from the research. The decision to give each portal its own primary accent colour (purple/teal/navy) was not pre-planned — it emerged when reviewing the personas and realising that Kavya (vendor) and Priya (customer) have entirely different mental models of what “this app” is. The colour becomes a psychological anchor.

Accessibility is a design constraint, not an afterthought. Addressing Vinod’s needs (larger text, transparent pricing, no dark patterns) and Rohit’s requirements (WCAG contrast, 60fps) early meant the design system was built accessible from day one rather than retrofitting it later.

Industry Best Practices Applied

- Atomic design system — components built once, reused everywhere
- Mobile-first approach — all 13 primary screens designed mobile before desktop
- Progressive disclosure — simple defaults with advanced options available on demand
- Jobs-to-be-done framing — each persona defined by goal, not demographics alone
- Zero-result and empty state design — every screen accounts for failure states

Future Roadmap

- **Phase 2:** AI-powered product recommendations based on browse history

- **Phase 2:** Vernacular language support (Hindi, Tamil, Marathi) — addresses Sunita persona
- **Phase 3:** Artisan/handmade curated section — “Made in India” editorial page (Meena persona)
- **Phase 3:** Vendor mobile app (separate from the web portal) with simplified inventory management
- **Phase 4:** Live commerce integration — vendor-hosted live product demonstrations
- **Phase 4:** Custom order workflow for artisan vendors (Meena’s biggest unmet need)

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